



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2015
Higher 2 Biology Paper 3 (9648/03) Answers

Answer **all** questions.

1

- (a)** Describe how this cDNA library can be created using plasmids. [5]

A cDNA library is made by first isolating all the mRNAs from a cell infected by the pathogen. Each of these mRNAs is then used as a template to make cDNA using reverse transcriptase enzyme. Mixture treated with alkali (NaOH)/ RNase to digest mRNA. The newly made cDNA strand is used as a template for DNA polymerase to create a complementary strand, producing a double stranded cDNA. Extend the 5' and 3' ends of the cDNA via the attachment of linkers. Restriction enzymes that produce sticky ends used to cut both plasmids and linker DNA. Mix the two types of restriction fragments to allow them to anneal by complementary base pairing. Incubate with DNA ligase to join the annealed strands covalently by forming phosphodiester bonds between adjacent nucleotides. Recombinant plasmids are introduced into competent bacterial cells. At least one colony in the cDNA library will carry a Rpi cDNA.

- (b) (i)** Explain how restriction enzymes recognize specific nucleotide sequences. [2]

The shape of the active site of these enzymes is complementary to the shape of the specific nucleotide sequences. These enzymes recognize and bind to the specific nucleotide sequences to form enzyme-substrate complexes.

- (ii)** Explain why the bacterial cells carrying the blight resistance gene can be identified using different selection media. [3]

EcoRI restriction site is located within the rifampicin resistance gene. Insertion of the gene into the plasmid disrupts the expression of rifampicin resistance gene. However, the other genetic marker kanamycin resistance gene is still intact. The transformed bacterial cells with recombinant plasmid will be able to grow in a medium containing only kanamycin, but NOT both rifampicin and kanamycin. Therefore by comparing the growth of bacteria on the master plate and replica plate, bacteria that survive in the medium containing kanamycin but not in medium containing both are the bacteria containing the gene of interest.

- (c) (i) Outline the technique of cloning new batches of transgenic plants from their explants. [4]

Surface disinfect the explant with bleach/ sodium hypochlorite solution. Place the explant in a sterile nutrient medium containing the following: a carbon and energy source, organic and inorganic nutrients, and plant growth regulators. The plant growth regulators will stimulate mitosis/ cell division and result in the formation of a callus from the cells placed in the medium. If the explant is successfully established, several microshoots are produced in a few weeks. The established and stabilized culture is transferred to a nutrient medium with an increased cytokinin and decreased auxin concentration to encourage shoot initiation. After the microshoots have reached an appropriate length, they are transferred to a root inducing medium where the increased auxin and reduced cytokinin concentration encourages root formation. The plantlet is shifted to the outdoor environment for acclimatization/ from a heterotrophic (sugar-requiring) to an autotrophic (free-living) condition in order for it to grow into a plant.

- (ii) Explain why these plants are described as clones. [2]

Ref. to Asexual reproduction AND plants come from one parent Ref. to genetically identical to one another and parents Plants share the same DNA/ same genotype/ same genome/ same alleles/ same number and types of chromosomes

2

- (a) Describe the effects of the mutated ADA gene on a SCID patient. [2]

Mutation in ADA gene leads to production of a non-functional enzyme adenosine deaminase (ADA) leading to accumulation of deoxyadenosine (dATP) which is toxic to lymphocytes. This affects the normal development of BOTH T- & B-lymphocytes. ADA-deficient children are born with severe immunodeficiency and are prone to repeated serious infections, which may be life-threatening.

- (b) (i) Suggest a suitable virus that can be used as a vector for gene therapy for long-term treatment of SCID. [1]

Retrovirus

- (ii) Explain briefly how a combination of gene therapy and stem cell therapy can lead to long-term treatment of SCID. [3]

The patient's blood stem cells are first isolated, and are treated with retrovirus that carry the normal functional ADA gene. These are modified ex vivo. Retroviruses also allow for integration of normal functional ADA gene into the genome of the blood stem cell. This ensures long-term treatment. These stem cells are then transplanted back to the patient. The genetically modified stem cells can differentiate into many T- and B-lymphocytes which will have a copy of the functional ADA gene. The genetically modified stem cells are capable of dividing and renewing themselves (self-renewal) for long periods. Therefore, there would be genetically modified stem cells in the patient's body for a long period of time. The use of the patients' own blood stem cells for gene therapy also prevents the immune system from destroying the cells. This also ensures that the treatment will be a long-term one.

- (c) With reference to Fig. 2.1 and 2.2, suggest how the complex achieves specific insertion of a normal gene into the host cell genome. [2]

RNA sequence on the guide mRNA can be modified so that it is complementary to a specific sequence of DNA in host cell genome near the site of insertion. Cas9 cutting protein hydrolyzes phosphodiester bonds between adjacent nucleotides so as to insert normal gene into a specific site on the genome.

- (d) Discuss **one** ethical issue surrounding somatic cell gene therapy. [2]

It is difficult to determine the risks/ long-term effects of exposure to viral vectors and the effects these engineered viruses have on the human genome. Even though every precaution is taken to prevent accidents, they sometimes do occur. Inadequate understanding of biological interactions of vector and host.

The attempts at gene therapy are extremely expensive. This could mean that only the very wealthy and powerful will have access to these therapies. There must be the establishment of procedural fairness in the selection of patients.

3

- (a) Explain why rice has been genetically modified to produce β -carotene. [2]

People in developing countries suffer from vitamin A deficiency. Rice is a staple diet in these countries. Golden rice provides people with β carotene which can form vitamin A / increases β -carotene in their diet / provide vitamin A in their diet. This can prevent blindness or reduces susceptibility to, diarrhoea, respiratory infections, measles and death.

- (b) Suggest **one** economical advantage of producing golden rice. [1]

Enable local farmers to grow to acquire economic gains from the sale of the crop

- (c) State **three** concerns about the use of GM crops.[3]

1. High cost of GM seeds,
2. Too much power held by multinational companies.

Risk to environment:

3. Gene transfer to wild-type relative or other related crops through cross-pollination. Changes to ecosystem through hybridisation.
4. Gene transfer from crop to crop via bacteria or viruses with unknown effect. Changes to ecosystem.
5. Recipient plant (plant that receives the gene) outcompetes natural variety or species / ref to 'superweed';;
6. Maybe toxic to non-pests;;
7. Unknown long term effects on the environment / biodiversity.

Risk to humans:

8. Allergy;;
9. Antibiotic resistance of gut bacteria if antibiotic marker used;;
10. Long term toxicity;;
11. Gene transfer from virus to human cells / gut bacteria, with unknown effect;;
12. No long term studies done on effects to human health.

- (d) Suggest why cDNA is used instead of *Psy* and *Crt1* genes. [2]

cDNA does not contain introns as they are reverse transcribed from mature mRNA. This helps to reduce the size of the recombinant plasmid to increase the chance that the rice endosperm cells will take up the recombinant plasmid successfully.

- (e) With reference to Fig. 3.1, deduce the sequence of the template strand. [1]

5' ATGGCACAGCGAACA 3'

- (f) State the sequence of the radioactive probe that could be used to locate the cDNA. [1]

3' TACCGTGTGCTTGT 5'

- (g) Other than the use of bacteria, state **one** method that can be used to introduce the recombinant plasmid into rice endosperm cells. [1]

Microinjection of rice endosperm cells/ electroporation/ virus

- (g) With reference to Table 3.1 and the information provided, discuss whether the hypothesis that the production of β -carotene in golden rice is limited by the source of the *Psy* gene is correct. [3]

The hypothesis is correct. *Psy* gene from daffodil only produced in 1a.u. of β -carotene while *Psy* gene from maize produces 14 a.u. of β -carotene, *psy* gene from pepper produces 4 a.u. of β -carotene and *psy* gene from tomato produces 6 a.u. of β -carotene. All recombinant plasmids have the same *Crt1* cDNA and an endosperm-specific promoter/ the only difference in the recombinant plasmids is the source of *Psy* gene.

4 Planning Question

In this experiment, the absorption spectra of two plant species, P and Q at different wavelengths of light are determined using a spectrophotometer by measuring the absorbance between 400 – 700 nm.

In the light-dependent reaction of photosynthesis, plant pigments will absorb light differentially at different wavelengths. Leaf extract can be prepared from the plants and then used to measure the amount of light absorbed. The spectrophotometer is adjusted to take the measurements at the different wavelengths, which are useful for photosynthesis/ photosynthetically active radiation.

The independent variable is wavelength of light (nm). The dependent variable is absorbance/ amount of light absorbed.

Apart from the wavelength of light, all other factors that can affect the amount of light absorbed should be kept constant as far as possible.

1. Pigment concentration

- Amount of photosynthetic pigments affects the absorption of light energy for photosynthesis.
- Use same number of leaves that are of the same size/mass to make fresh leaf extract to keep pigment concentration constant.

2. Volume of extraction buffer used

- It affects the amount of pigments, thus the absorbance value of the solution.
- Use a 10 ml syringe to dispense 10 ml of buffer in each test tube to keep the volume of solution constant.

PROCEDURE

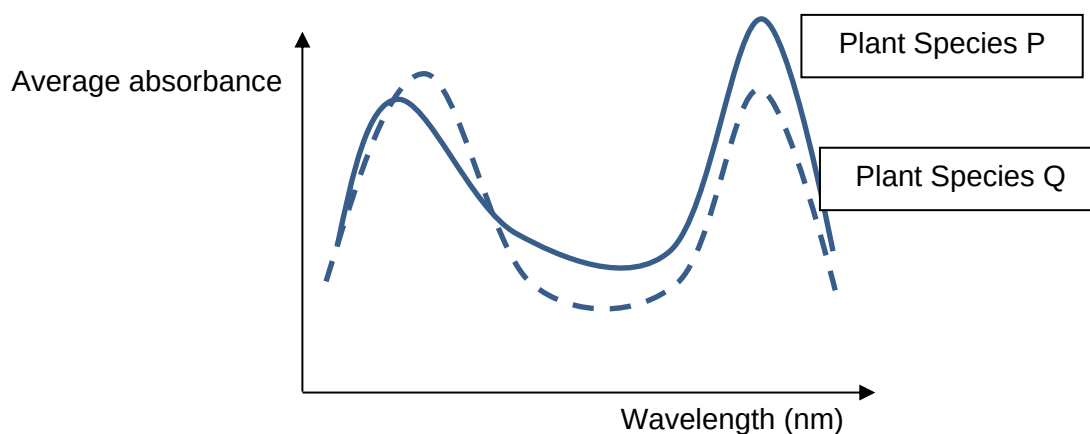
1. Dispense all solutions separately with clean and dry syringes each time and start with clean and dry glassware (e.g. capillary tube) for each reading and repeat to prevent contamination.
2. Using a sharp knife, discard large veins and cut the leaves of plant species P into smaller pieces.
3. Weigh and place 2g (fixed mass) of cut leaves into a mortar. Add sand and use a syringe to add 12ml of the pigment extraction buffer.
4. Grind leaves with pestle for 2 minutes to release the chloroplasts.
5. Slowly pour the contents of the mortar onto a filter funnel lined with muslin cloth.
6. Allow the mixture to stand for a few minutes to collect the leaf extract/filtrate.
7. Repeat Steps 2 to 6 for plant species Q.
8. Pour the leaf extract from plant species P into a clean cuvette.
9. Adjust the wavelength on the spectrophotometer at 400 nm and record the absorbance reading of the leaf extract.
10. Take the absorbance readings of the same leaf extract at 450, 500, 550, 600, 650 and 700 nm.
11. Remove the cuvette from the spectrophotometer and fill another clean cuvette with the leaf extract from plant species Q.
12. Repeat steps 9 and 10.

13. To ensure reliability of results, repeat steps 9 to 13 to obtain a total of 3 readings (triplicates) using fresh samples of leaf extract prepared from the same batch of leaves.
14. To ensure reproducibility of data, repeat the entire experiment (steps 1 to 15) twice using freshly prepared leaf extracts from different batches of leaves for both species.

Tables showing absorbance of different plant species at different wavelengths of light (nm)

Wavelength (nm)	Absorbance of Plant Species P				Absorbance of Plant Species Q			
	Trial 1	Trial 2	Trial 3	Ave	Trial 1	Trial 2	Trial 3	Ave
400								
450								
500								
550								
600								
650								
700								

Graph of average absorbance of plant species P and Q against wavelength of light (nm)



As different plant species have different compositions of pigments, they will have different absorption spectra. The absorption spectra will show the wavelengths that the pigments absorb the most or least light.

SAFETY PRECAUTIONS

1. The pigment extraction buffer is an irritant therefore wear goggles and gloves, and wash under running water upon contact with the skin or eye.
2. Handle the sharp knife with care to avoid being cut.
3. Use the mortar and pestle carefully to prevent injuries/ accidental pounding of fingers.

5 Free-response question

- (a) Explain the advantages and limitations of polymerase chain reaction (PCR). [8]

Advantages:

Fast and easy to use; a typical PCR run can be completed in less than two hours (cloning into a vector requires several days) and the process is automated. This is because DNA denaturation and annealing of primers are controlled by temperature and extension of primers is carried out Taq polymerase, which is thermostable.

PCR targets only the desired DNA region to be copied through the use of forward and reverse primers, which ensures only the DNA segment of interest will be amplified.

Limitations:

Only short sequences can be accurately amplified. The longer the sequence, the higher the rate of error because Taq polymerase does not have proofreading ability.

Sequences flanking the DNA segment of interest have to be known for the design of forward and reverse primers so that only the DNA segment of interest will be amplified

Any DNA template with complementary sequences to the forward and reverse primer will be amplified. Unwanted DNA segments may be amplified, especially when there is DNA from other sources.

- (b) Explain what is meant by restriction fragment length polymorphism (RFLP) and how it can be detected. [6]

RFLP: restriction fragments of different lengths when digested by the same type of restriction enzyme. This is due to differences in nucleotide sequence between alleles / between genetic markers at a chromosomal locus. As a result, different numbers of restriction enzyme sites or different numbers of tandem repeats (VNTR = variable number tandem repeat / STR = short tandem repeats) is observed.

It can be detected by Southern Blotting, where the chromosomal locus is digested then with restriction enzyme. The restriction fragments produced are separated using gel electrophoresis and transferred to a nylon membrane. DNA fragments from this locus are identified using a radioactive probe which produces a dark band on the photographic film. Polymorphism would result in different banding patterns.

(c) Discuss the aims of the human genome project.

[6]

To determine the sequences of all the 3.2 billion base pairs in the human genome with a minimal error rate, so as to identify all the 20,000 to 25,000 genes in the human DNA and to construct a genetic linkage map and physical map of the human genome. This would enable the development of genetic tests/screening for diseases and develop more efficient methods for DNA sequencing and sequence analysis and the transfer of these technologies to industry.

Human genome information would be managed through storage of the sequence of the human DNA in databases and making it accessible to anyone on the Internet. Computer programs or software would be developed and improved to analyze the data because the data are difficult to interpret without such programs. This would also developed transfer related technologies (DNA sequencing, PCR etc) to enhance and assist scientific research.

However, there are uncertainty issues associated with disease susceptibilities and complex conditions identified by genetic testing. It may also lead to genetic discrimination.

[Total: 20]

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